Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Conjecture : Convergence of Collatz (3n+1) Sequence to the Trivial Cycle

A DEFINITIVE PROOF & HALEMANE-CONJECTURE

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The entire dynamics of the Collatz (3n+1) system can be exactly represented by a dynamically-evolving graded-algebraic-structure of an ideal-based-filtration-scheme; with divisibility-classes for modulo-3 quotient-semiring generated by the primitive root 2; along with a filtration-shifting-global-affine-transformation, f(x)=(3x+1) when x is a positive odd number; achieving a Euclidean expansion shift to the topmost coprime-layer of that filtration-scheme; while also avoiding the modulo-multiple-layer (zero-layer) and all the nilpotent-layers with nilpotent-elements (dead-end zero-divisors) like (6m-3) and (6m-0); the trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$ being bypassed by an initialization-phase for this filtration scheme, starting directly with the modul-9 coprime-layer. The transitive closure of the Collatz (3n+1) system is exactly represented by the transitive closure of this dynamically evolving algebraic system which is indeed the complete set of natural numbers. This design for the system-dynamics-framework by itself provides a definitive-proof of the Collatz (3n+1) conjecture, establishing that the entire Collatz map is exactly represented by this dynamically-evolving graded-algebraic-structure defined on the set of all natural numbers; neither missing any natural-number nor including any extraneous-elements; providing both the necessary-&-sufficient condition simultaneously in one single sweep.

REFER: https://archive.org/download/CTUHSK/CTUHSK.pdf

Halemane-Conjecture states that the maximum number of odd (3n+1) operations required to reach the trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$ starting from any given positive integer and moving along the Collatz sequence, is limited by that given number itself; with the triad $\{(31 \Leftarrow 41 \Leftarrow 27)\}$ as an exceptional limiting case.